

Lecture 4: Itô's Lemma, SDEs, and Feynman-Kac

Lecturer: Andrew Papanicolaou

Scribes: Daniel Chertock

Note: *LaTeX template courtesy of UC Berkeley EECS department.*

Disclaimer: *This note has not been subjected to formal publication review. Distribution outside class requires instructor permission.*

4.1 2nd-Order Taylor Series and The Box Rules

The derivation of Itô's lemma follows from our understanding of this Itô integral. Itô's lemma is the analogue of the derivative as it was derived in the previous lecture, except there is a 2nd-order term that remains. Itô's lemma is essential for handling and solving of stochastic differential equations (SDEs).

Let W_t be a standard Brownian motion. Let g be bounded and differentiable function, and let f be a three-times continuously differentiable function with $f'''(w)$ bounded. Let $\{t_i\}_{i=0}^n$ be a shrinking partition of $[0, t]$ as $n \rightarrow \infty$. A general stochastic integral is

$$I_t[g] = \int_0^t g(W_t) df(W_t) = \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} g(W_i) \Delta f(W_i), \quad (4.1)$$

where $W_i = W_{t_i}$ and $\Delta f(W_i) = f(W_{t_{i+1}}) - f(W_{t_i})$. The increment $\Delta f(W_i)$ can be broken down with a Taylor expansion,

$$\Delta f(W_i) = f'(W_i) \Delta W_i + \frac{1}{2} f''(W_i) \Delta W_i^2 + \frac{1}{3!} f'''(\xi_i) \Delta W_i^3,$$

where ξ_i is some random intermediate point between W_i and W_{i+1} that makes the expansion exact by the mean-value theorem. Since W_i is known at time t_i , the quantities evaluated at W_i do not anticipate the future Brownian increment $\Delta W_i = W_{t_{i+1}} - W_{t_i}$. The intermediate point ξ_i , however, may depend on this increment. We use this expansion to rewrite the summation in (4.1),

$$I_t[g] = \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} g(W_i) \left(f'(W_i) \Delta W_i + \frac{1}{2} f''(W_i) \Delta W_i^2 + \frac{1}{3!} f'''(\xi_i) \Delta W_i^3 \right).$$

There are three terms in the above expression:

$$\lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} g(W_i) f'(W_i) \Delta W_i = \int_0^t g(W_t) f'(W_t) dW_t \quad (4.2)$$

$$\lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} g(W_i) f''(W_i) \Delta W_i^2 = \int_0^t g(W_t) f''(W_t) dt \quad (4.3)$$

$$\lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} g(W_i) f'''(\xi_i) \Delta W_i^3 = 0, \quad (4.4)$$

where these limits are in the sense of mean square. The limit shown in (4.2) is the very definition of the Itô integral from our previous lecture. For $\Delta t = t/n$, the limit shown in (4.3) is easy to show. Denote the bound $|g(w)f''(w)|^2 \leq C$ for all w , in which case

$$\begin{aligned}
& \mathbb{E} \left(\sum_{i=0}^{n-1} g(W_i) f''(W_i) (\Delta W_i^2 - \Delta t) \right)^2 \\
&= \sum_{i=0}^{n-1} \mathbb{E} \left(g(W_i) f''(W_i) (\Delta W_i^2 - \Delta t) \right)^2 \\
&\leq C \sum_{i=0}^{n-1} \mathbb{E} (\Delta W_i^2 - \Delta t)^2 \\
&= nC (\mathbb{E} \Delta W_i^4 - 2\Delta t \mathbb{E} \Delta W_i^2 + \Delta t^2) \\
&= nC (3\Delta t^2 - 2\Delta t^2 + \Delta t^2) \\
&= 2nC \Delta t^2 \\
&= \frac{2Ct^2}{n} \\
&\rightarrow 0
\end{aligned}$$

as $n \rightarrow \infty$. Similar, for $|g(w)f'''(w')|^2 \leq C$ for all w and w' , then the limit in (4.4) also tends to zero,

$$\begin{aligned}
& \mathbb{E} \left(\sum_{i=0}^{n-1} g(W_i) f'''(\xi_i) \Delta W_i^3 \right)^2 \\
&= \sum_{i=0}^{n-1} \mathbb{E} \left(g(W_i) f'''(\xi_i) \Delta W_i^3 \right)^2 \\
&\leq nC \mathbb{E} \Delta W_i^6 \\
&= 15nC \Delta t^3 \\
&= \frac{15Ct^3}{n^2} \\
&\rightarrow 0,
\end{aligned}$$

as $n \rightarrow \infty$. These limits prove to us the limits of (4.3) and (4.4), and give use the box rules,

$$\begin{aligned}
dW_t^2 &= dt \\
dW_t dt &= 0 \\
dt^2 &= 0,
\end{aligned}$$

from which it follows that all other higher-order increments can be ignored. In the multi-variate case we have the following additional rule:

$$dW_t dW'_t = 0,$$

where W_t and W'_t are independent Brownian motions. Using these box rules, the stochastic integral in (4.1) can be expressed with f' and f'' ,

$$I_t[g] = \int_0^t g(W_s) df(W_s) = \int_0^t g(W_s) f'(W_s) dW_s + \frac{1}{2} \int_0^t g(W_s) f''(W_s) ds. \quad (4.5)$$

4.2 Itô's Lemma

From (4.5) we can deduce the stochastic differential for $f(W_t)$, effectively giving us Itô's lemma when the underlying process is standard Brownian motion,

$$df(W_t) = f'(W_t)dW_t + \frac{1}{2}f''(W_t)dt . \quad (4.6)$$

Itô's lemma as it's given in (4.6) can be generalized for stochastic processes of the form

$$X_t = X_0 + \int_0^t a(t, X_t)dt + \int_0^t \sigma(t, X_t)dW_t ,$$

Or in differential form we can express X_t as stochastic differential equation (SDE)

$$dX_t = a(t, X_t)dt + \sigma(t, X_t)dW_t ,$$

with initial condition X_0 . Using a Taylor expansion and the box rules, the Itô lemma in (4.6) can be generalized to $f(t, X_t)$, which is how it is commonly written,

$$df(t, X_t) = \left(f_t(t, X_t) + a(t, X_t)f_x(t, X_t) + \frac{1}{2}\sigma^2(t, X_t)f_{xx}(t, X_t) \right) dt + \sigma(t, X_t)f_x(t, X_t)dW_t . \quad (4.7)$$

Itô's lemma is a powerful tool and is the key to solving many problems in continuous-time finance; if you can apply Itô's lemma then you can make progress toward a solution to a problem.

4.3 Existence and Uniqueness of SDEs

Before exploring explicit solutions, it is crucial to specify what is meant by a strong solution to an SDE. Consider the general SDE

$$dX_t = a(t, X_t)dt + \sigma(t, X_t)dW_t ,$$

with initial condition X_0 . A **strong solution** is a continuous stochastic process X_t that satisfies the SDE with probability one, is adapted to the filtration generated by W_t , and is constructed from the given path of the Brownian motion.

A solution is **square integrable** if

$$\mathbb{E} \left[\int_0^T |X_t|^2 dt \right] < \infty .$$

A square-integrable strong solution exists and is unique if the drift coefficient $a(t, x)$ and diffusion coefficient $\sigma(t, x)$ satisfy two key conditions for all $x, y \in \mathbb{R}$ and $t \in [0, T]$:

1. **Lipschitz Continuity:** There exists a constant K such that

$$|a(t, x) - a(t, y)| + |\sigma(t, x) - \sigma(t, y)| \leq K|x - y| .$$

2. **Linear Growth:** There exists a constant C such that

$$|a(t, x)| + |\sigma(t, x)| \leq C(1 + |x|) .$$

Equivalently, a squared version is often useful:

$$|a(t, x)|^2 + |\sigma(t, x)|^2 \leq C(1 + |x|^2) ,$$

since it is convenient for showing that the solution's second moment does not grow too quickly.

The Lipschitz condition prevents the solution from branching (ensuring uniqueness), while the linear growth bound ensures the solution does not explode to infinity in finite time (ensuring existence).

4.4 Examples

Example 1 (Arithmetic Brownian Motion). Let X_t be an arithmetic Brownian motion $X_t = X_0 + at + \sigma W_t$. For $f(x) = e^x$ let's apply Itô's lemma given by (4.7) to $f(X_t)$,

$$df(X_t) = \left(a + \frac{\sigma^2}{2}\right) f(X_t)dt + \sigma f(X_t)dW_t .$$

We can define $Y_t = f(X_t)$, for which we have an SDE,

$$\frac{dY_t}{Y_t} = \left(a + \frac{\sigma^2}{2}\right) dt + \sigma dW_t .$$

Example 2 (Geometric Brownian Motion). Let S_t have the SDE

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t .$$

Letting $X_t = \log(S_t)$, we apply Itô's lemma,

$$dX_t = \frac{dS_t}{S_t} - \frac{(dS_t)^2}{2S_t^2} = \left(\mu - \frac{\sigma^2}{2}\right) dt + \sigma dW_t .$$

Hence, we have a solution to the SDE,

$$S_t = e^{X_t} = S_0 e^{\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W_t} .$$

The process S_t is called Geometric Brownian motion, and is log-normally distributed.

Example 3 (Ornstein-Uhlenbeck Process). Consider X_t an Ornstein-Uhlenbeck (OU) process,

$$dX_t = \lambda(\alpha - X_t)dt + \sigma dW_t ,$$

with $\lambda > 0$ so that X_t is mean-reverting to α . We can use the integrating factor $e^{\lambda t}$ to solve the SDE. Set $Y_t = f(t, X_t) = e^{\lambda t} X_t$ and apply Itô's lemma given by (4.7),

$$dY_t = \lambda Y_t dt + e^{\lambda t} dX_t = \alpha \lambda e^{\lambda t} dt + \sigma e^{\lambda t} dW_t ,$$

which integrates to

$$Y_t = Y_0 + \alpha \lambda \int_0^t e^{\lambda s} ds + \sigma \int_0^t e^{\lambda s} dW_s .$$

Now multiply both side by $e^{-\lambda t}$ for the solution of the SDE for X_t ,

$$X_t = e^{-\lambda t} X_0 + \alpha (1 - e^{-\lambda t}) + \sigma \int_0^t e^{-\lambda(t-s)} dW_s .$$

The stochastic integral in this solution is normally distributed,

$$\int_0^t e^{-\lambda(t-s)} dW_s \sim \text{normal}\left(0, \frac{1 - e^{-2\lambda t}}{2\lambda}\right) .$$

Example 4 (Non-Normality of $\int_0^t W_s dW_s$). An example of a non-normal stochastic integral is $\int_0^t W_s dW_s$. We can apply Itô's lemma to see that this integral's distribution is expressed with a chi-squared random variable. First, apply Itô's lemma to W_t^2 ,

$$dW_t^2 = 2W_t dW_t + dt .$$

Then see that perform a generalization of integration by parts for stochastic integrals,

$$\int_0^t W_s dW_s = \frac{1}{2} W_t^2 - \frac{t}{2} ,$$

which has a non-central chi-squared distribution equal to $\frac{t}{2} (\chi_1^2 - 1)$ where χ_1^2 is chi-squared distributed with 1 degree of freedom.

Example 5 (Multi-Variate Itô's Lemma). Let W_t^1 and W_t^2 be two Wiener processes, and consider the following SDEs,

$$\begin{aligned} dX_t &= a(X_t)dt + b(X_t)dW_t^1 \\ dY_t &= \mu(X_t)dt + \sigma(X_t)dW_t^2 . \end{aligned}$$

For any function $f(x, y)$ denote $Z_t = f(X_t, Y_t)$. If W_t^1 and W_t^2 are independent,

$$dZ_t = f_x dX_t + f_y dY_t + \frac{1}{2} (b^2 f_{xx} + \sigma^2 f_{yy}) dt .$$

If W_t^1 and W_t^2 have correlation coefficient $\rho \in [-1, 1]$, then there is an extra term,

$$dZ_t = f_x dX_t + f_y dY_t + \left(\frac{1}{2} b^2 f_{xx} + \frac{1}{2} \sigma^2 f_{yy} + \rho b \sigma f_{xy} \right) dt .$$

4.5 Dynkin's Formula and the Feynman-Kac Theorem

The true power of Itô's lemma becomes evident when bridging SDEs, expected values, and partial differential equations (PDEs). Let X_t be an Itô diffusion satisfying $dX_t = a(t, X_t)dt + \sigma(t, X_t)dW_t$. We define the infinitesimal generator $\mathcal{L}$ acting on a sufficiently smooth function $f(x)$ as

$$\mathcal{L}f(x) = \frac{1}{2} \sigma^2(t, x) \frac{\partial^2 f}{\partial x^2} + a(t, x) \frac{\partial f}{\partial x} .$$

Dynkin's Formula provides the expected value of the process at time t , acting as a stochastic generalization of the Fundamental Theorem of Calculus:

$$\mathbb{E}_x[f(X_t)] = f(x) + \mathbb{E}_x \left[\int_0^t \mathcal{L}f(X_s) ds \right] ,$$

where $\mathbb{E}_x$ denotes the expectation given the initial state $X_0 = x$.

Building upon this connection is the **Feynman-Kac Theorem**, which provides a probabilistic solution to deterministic parabolic PDEs. Suppose a function $u(t, x)$ solves the boundary value problem

$$\begin{aligned} \frac{\partial u}{\partial t} + a(t, x) \frac{\partial u}{\partial x} + \frac{1}{2} \sigma^2(t, x) \frac{\partial^2 u}{\partial x^2} - r(t, x)u &= 0 , \quad (t, x) \in [0, T) \times \mathbb{R} \\ u(T, x) &= \Phi(x) , \end{aligned}$$

where $r(t, x) \geq 0$ is a potential function (often representing a discount rate). The Feynman-Kac representation implies the solution can be written as a conditional expectation:

$$u(t, x) = \mathbb{E} \left[e^{-\int_t^T r(s, X_s) ds} \Phi(X_T) \middle| X_t = x \right] .$$

This theorem is foundational in modern continuous-time finance, demonstrating that solving pricing PDEs (like the Black-Scholes equation) is mathematically equivalent to computing an expected discounted payoff under a risk-neutral measure.

Example 6 (Black-Scholes and Feynman-Kac). *Consider the pricing of a European derivative with a terminal payoff $\Phi(S_T)$ at maturity T . The price of the derivative $V(t, s)$ is known to satisfy the Black-Scholes PDE:*

$$\frac{\partial V}{\partial t} + rs \frac{\partial V}{\partial s} + \frac{1}{2} \sigma^2 s^2 \frac{\partial^2 V}{\partial s^2} - rV = 0 ,$$

with the terminal condition $V(T, s) = \Phi(s)$.

We can map this directly to the Feynman-Kac theorem by identifying our variables: the drift is $a(t, s) = rs$, the diffusion is $\sigma(t, s) = \sigma s$, and the continuous discount rate is a constant r .

Applying the theorem, the solution to the PDE is given by the conditional expectation:

$$V(t, s) = \mathbb{E} \left[e^{-r(T-t)} \Phi(S_T) \middle| S_t = s \right] ,$$

where the underlying asset S_t evolves according to the associated SDE (the risk-neutral dynamics):

$$dS_t = rS_t dt + \sigma S_t dW_t .$$

This elegantly demonstrates that solving the Black-Scholes PDE is mathematically equivalent to calculating the expected discounted payoff of the derivative.

4.6 Remarks

1. A good background reference for this material is [1] and [2].

References

- [1] T. Björk. *Arbitrage Theory in Continuous Time*. Oxford University Press, Oxford, UK, 4th edition, 2020.
- [2] B. Oksendal. *Stochastic differential equations: an introduction with applications*. Springer Science & Business Media, 2013.